protopus

CIC-TBI · QUARTERLY PROGRESS REVIEW · SPEAKER SCRIPT

Ten minutes, ten slides.

What to say on each slide, with timings. Speak slowly; the timings leave room for that. If time is short, shorten slides 03 to 05 (July, August, September) and keep the rest.

#	SLIDE	TIME	LENGTH
00	Cover	0:00 – 0:20	20 s
01	What we are building	0:20 – 1:20	60 s
02	The quarter at a glance	1:20 – 2:20	60 s
03	July	2:20 – 3:50	90 s
04	August	3:50 – 5:20	90 s
05	September: integration and the business	5:20 – 6:50	90 s
06	September: funding, Probe Matrix, plan	6:50 – 7:50	60 s
07	The quarter in numbers	7:50 – 8:30	40 s
08	Next quarter	8:30 – 9:40	70 s
09	Thank you	9:40 – 10:00	20 s
	Total		**10 min**

SLIDE 00 · 0:00 – 0:20

Cover

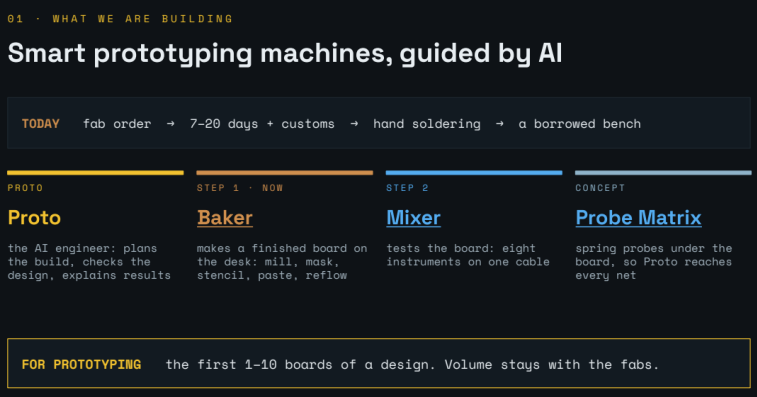


SAY (ABOUT 20 SECONDS)

Good afternoon, sir. I am Soham, founder of Protopus. Thank you for your patience today. In the next ten minutes I will take you through what we did from July to September, month by month, and what we plan to do next.

SLIDE 01 · 0:20 – 1:20

What we are building

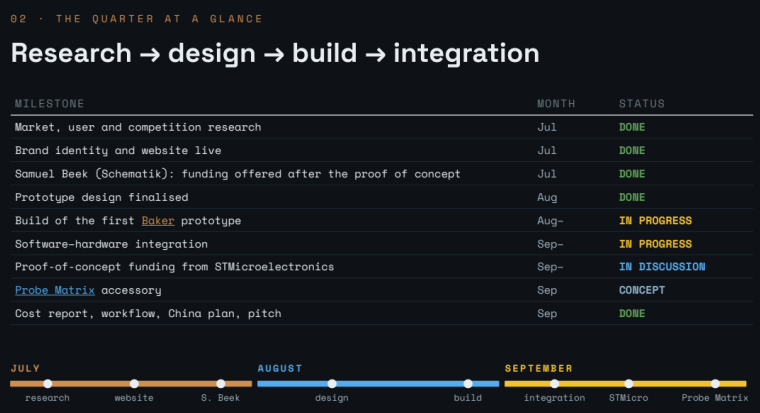


SAY (ABOUT 60 SECONDS)

First, a quick reminder of what Protopus is. Today, a first circuit board in India means a fab order, seven to twenty days of shipping and customs, hand soldering, and a borrowed test bench. AI has made software easy to build, but not hardware, because AI has no hands in the real world. We build those hands. Proto is our AI engineer: it plans the build. Baker, our first machine, makes a finished board on the desk. Mixer, the second, will test it. The Probe Matrix is a new accessory that lets Proto reach every point on the board. And we are clear about one thing: this is for prototyping, the first one to ten boards of a design. Volume stays with the fabs.

SLIDE 02 · 1:20 – 2:20

The quarter at a glance



02 · THE QUARTER AT A GLANCE

Research → design → build → integration

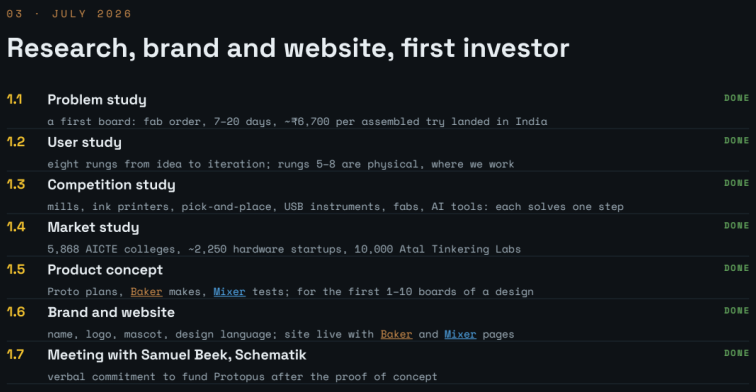
MILESTONE	MONTH	STATUS
Market, user and competition research	Jul	DONE
Brand identity and website live	Jul	DONE
Samuel Beek (Schematik): funding offered after the proof of concept	Jul	DONE
Prototype design finalised	Aug	DONE
Build of the first Baker prototype	Aug–	IN PROGRESS
Software-hardware integration	Sep–	IN PROGRESS
Proof-of-concept funding from STMicroelectronics	Sep–	IN DISCUSSION
Probe Matrix accessory	Sep	CONCEPT
Cost report, workflow, China plan, pitch	Sep	DONE

SAY (ABOUT 60 SECONDS)

Here is the quarter on one page. In July we finished our research, launched the website and had our first investor conversation. In August we finalised the prototype design and started building it. In September we worked on connecting the software to the hardware, opened funding talks for the proof of concept, and conceived the Probe Matrix. The colours show where each item stands: green is done, yellow is in progress, blue is in discussion. I have only marked something done when it is actually done.

SLIDE 03 · 2:20 – 3:50

July



03 · JULY 2026

Research, brand and website, first investor

1.1	**Problem study** a first board: fab order, 7–20 days, ~₹6,700 per assembled try landed in India	DONE
1.2	**User study** eight rungs from idea to iteration; rungs 5–8 are physical, where we work	DONE
1.3	**Competition study** mills, ink printers, pick-and-place, USB instruments, fabs, AI tools: each solves one step	DONE
1.4	**Market study** 5,868 AICTE colleges, ~2,250 hardware startups, 10,000 Atal Tinkering Labs	DONE
1.5	**Product concept** Proto plans, Baker makes, Mixer tests; for the first 1–10 boards of a design	DONE
1.6	**Brand and website** name, logo, mascot, design language; site live with Baker and Mixer pages	DONE
1.7	**Meeting with Samuel Beek, Schematik** verbal commitment to fund Protopus after the proof of concept	DONE

SAY (ABOUT 90 SECONDS)

July was research and our first step outside. We mapped how a first board is made today: one assembled try from JLCPCB, landed in India, costs about six thousand seven hundred rupees and takes up to twenty days. We studied where people get stuck, in eight rungs from idea to iteration, and saw that the first four are being solved by AI software, while the last four are physical. That is where we chose to work. We studied every competitor and their prices; each one solves only one step. We sized our first market in India: about fifty-nine hundred engineering colleges, the hardware startups, and ten thousand Atal Tinkering Labs. Then we defined the product, built the Protopus brand, and launched the website with pages for Baker and Mixer. At the end of July we met Samuel Beek, founder and CEO of Schematik. He gave us a verbal commitment to fund Protopus once the proof of concept is achieved.

SLIDE 04 · 3:50 – 5:20

August

04 · AUGUST 2026

Prototype design finalised; build started

2.1	Baker 15-step process feed, probe, mill, drill, mask, stencil, paste, jig, reflow, test	DONE
2.2	Machine architecture one shuttle, two zones, one coordinate frame; refill magazine	DONE
2.3	Subsystem specification 500 W spindle, 10 W laser, 405 nm LCD, 138 °C reflow; Pi 5 + Klipper	DONE
2.4	Mixer architecture eight instruments on one FPGA; for after the Baker proof of concept	DONE
2.5	Parts and bill of materials every subsystem selected, so procurement could start	DONE
2.6	Build of the real prototype procurement and assembly of the first Baker	IN PROGRESS



Baker, render of the design

SAY (ABOUT 90 SECONDS)

In August we turned the idea into an engineering design. We fixed Baker's fifteen-step process: feed and probe the board, mill and drill it, apply and cure the solder mask, cut the stencil and paste, place the parts using a printed jig, reflow, and hand it over for test. We chose one moving shuttle that carries the board through two zones in a single coordinate frame, with a refill magazine so a board does not need loading each time. We specified every subsystem: the spindle, the laser, the UV screen for the mask, the low-temperature reflow, and the control electronics running Klipper. We also defined Mixer's architecture for a later stage. Then we selected the parts, built the bill of materials, and in late August we started building the first Baker. The build is in progress.

SLIDE 05 · 5:20 – 6:50

September: integration and the business

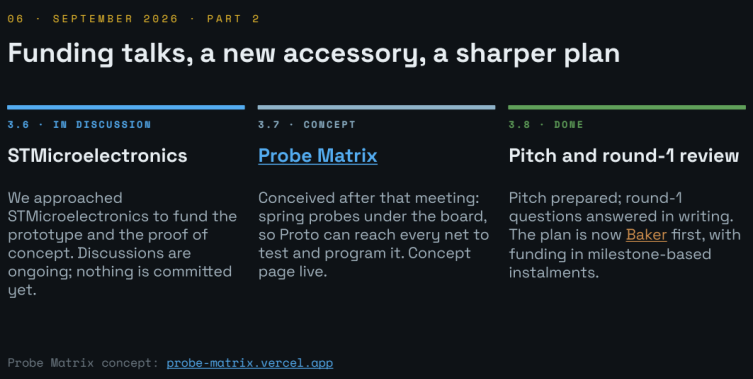


SAY (ABOUT 90 SECONDS)

September had two halves. The first was software and documentation. We worked on Baker Studio, where Proto reads a design through the Flux AI interface, flags problems like a trace that is too thin, proposes the fix, and plans the machine job. We are connecting it to the machine through the Klipper firmware; that integration is in progress. We costed the full bill of materials, line by line. The first Baker prototype comes to about three lakh seventy-three thousand rupees, including duty and a fifteen percent contingency. At a hundred units, with parts from China, production is projected at about one lakh forty-five thousand. We also wrote the workflow document, which follows one board through all fifteen steps, and a manufacturing plan: turnkey production in China, with final testing, minimal assembly and shipping from India.

SLIDE 06 · 6:50 – 7:50

September: funding, Probe Matrix, plan

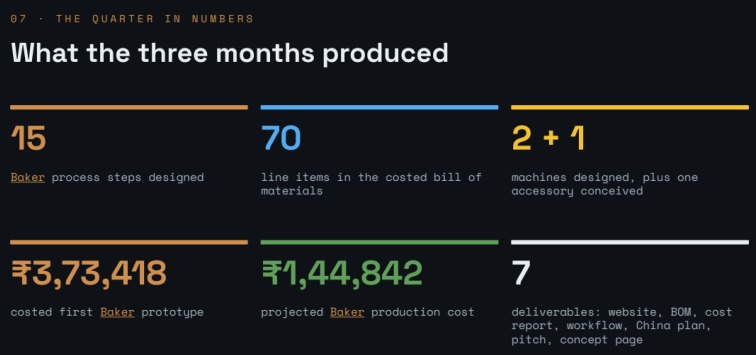


SAY (ABOUT 60 SECONDS)

The second half was about the road ahead. We approached STMicroelectronics to fund the prototype and the proof of concept. Those discussions are ongoing; nothing is committed yet. After that meeting we conceived the Probe Matrix: spring probes under the board, so Proto can reach every point to test and program it without a person moving probes. The concept page is live. Finally, we prepared our pitch and received round-one review questions. As a result we are going Baker first, one product at a time, with funding released in milestone-based instalments.

SLIDE 07 · 7:50 – 8:30

The quarter in numbers

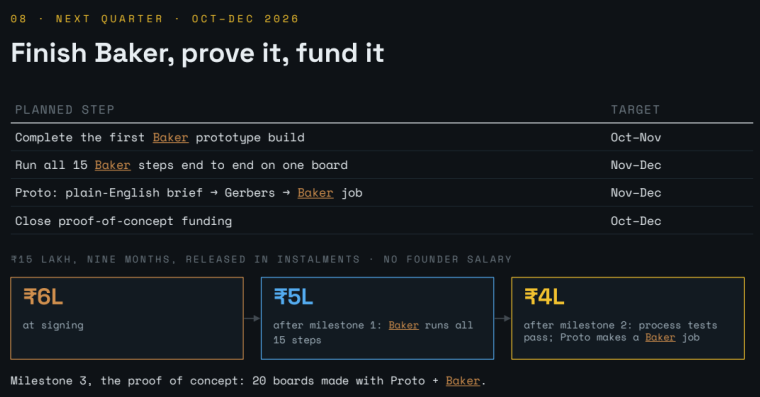


SAY (ABOUT 40 SECONDS)

In numbers: a fifteen-step process designed, a seventy-line costed bill of materials, two machines designed and one accessory conceived, a costed prototype of about three point seven three lakh, a projected production cost of about one point four five lakh, and seven deliverables completed, from the website to the concept page.

SLIDE 08 · 8:30 – 9:40

Next quarter



SAY (ABOUT 70 SECONDS)

For October to December, our focus is the Baker prototype. We will complete the build, run all fifteen steps end to end on one board, take Proto from a plain-English brief to a Baker job, and close the proof-of-concept funding. We are seeking fifteen lakh rupees for a nine-month proof of concept, released in three instalments: six lakh at signing, five lakh once Baker runs all fifteen steps, and four lakh once the process tests pass and Proto produces a Baker job. The proof of concept is twenty boards made with Proto and Baker. No founder salary is drawn.

SLIDE 09 · 9:40 – 10:00

Thank you

protopus

Thank you, sir.

Questions and guidance welcome.

WEBSITE	https://protopus-kappa.vercel.app/
BAKER	https://protopus-kappa.vercel.app/#/baker
MIXER	https://protopus-kappa.vercel.app/#/mixer
PROBE MATRIX	https://probe-matrix.vercel.app/

Soham · Founder, Protopus · soham9763015@ece.du.ac.in



SAY (ABOUT 20 SECONDS)

That is our quarter, sir. The report has the full details, and the links are on the screen. Thank you. I would be glad to take your questions and your guidance for the next quarter.

BE READY FOR

Questions you may be asked, and short honest answers

When will the prototype run end to end?

We are completing the build in October and November, and our target is to run all 15 steps on one board in November to December.

What exactly has Samuel Beek committed?

A verbal commitment to fund Protopus once the proof of concept is achieved. Nothing is signed yet; the proof of concept is the trigger.

Where are you with STMicroelectronics?

We have approached them for proof-of-concept funding and discussions are ongoing. Nothing is committed yet.

Why Baker first, and not both machines?

One product at a time. Baker proves the hardest part, making the board. Mixer and the Probe Matrix follow after the proof of concept.

Isn't this just a desktop PCB mill?

No. A mill does one step for experts. Baker runs 15 steps in one machine, from bare board to reflowed board, and Proto guides a first-timer through each one.

Why ₹15 lakh, and why in instalments?

It covers the Baker build, Rev B, one engineer for nine months, compute, testing and IP, with no founder salary. Instalments tie each release to a milestone you can see.

What do you need from CIC?

Prepare your own answer before the meeting, for example lab space, mentoring, or introductions to investors or industry partners.